

Naveen Kumar Gupta

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MS(R)
Electrical Engineering
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Education

Degree/Certificate	Institute/Board	CGPA/Percentage	Year
MS(R), EE	Indian Institute of Technology Indore	(Current)	2026-Present
B.Tech. ECE	Indian Institute of Information Technology Ranchi	9.58	2022-26
Senior Secondary	Sunbeam Bhagwanpur, Varanasi, CBSE Board	90.8%	2021
Secondary	St. Joseph's, Chandauli, CBSE Board	91.8%	2019

Experience

- **SPWiCOM, Indian Institute of Science (IISc)** *Mar. 2026 - Jun. 2026*
  Research Intern Bengaluru
  – Developed a passive radar processing pipeline, separating 2 non-cooperative transmitters and achieving over 40 dB interference suppression to detect a target moving at 20 m/s at 15 km range.
- **Amity University** *Jun. 2025 - Jul. 2025*
  Undergraduate Summer Research Fellow Noida
  – Designed a reversible quantum adder-subtractor using Qiskit and optimized PERES/CNOT gate usage. Implemented complementor-based result validation and exhaustively verified all 32 input combinations across 2-bit addition and subtraction modes.
- **Netaji Subhas University of Technology (NSUT)** *May. 2024 - Jul. 2024*
  Undergraduate Summer Research Fellow Delhi
  – Analyzed optical waveguides using COMSOL Multiphysics, simulating 2 SOI resonator types—ring and racetrack. Optimized grating-integrated structures and geometric parameters to enhance resonator performance and achieve targeted optical behavior.

Projects

- **MIPS 32-bit Pipelined Processor** *Jul. 2026*
  Verilog HDL
  – Implemented a 32-bit MIPS processor in Verilog with a 5-stage pipeline (IF–ID–EX–MEM–WB), 13 instructions, a 32×32-bit register file, and 1024×32-bit memory;
  – Verified arithmetic, logical, load/store, and branch operations using simulation-based testbenches, achieving up to 4×throughput improvement over single-cycle execution.
  – Extending the processor architecture to develop a **hardware accelerator** for Edge AI applications, targeting efficient computation through optimized multiply-accumulate (MAC) operations and pipelined CORDIC-based multiplication.
- **SystemVerilog FIFO Verification Environment** *Nov. 2025*
  SystemVerilog
  – Developed a constrained-random, self-checking SystemVerilog testbench for a 16×8 asynchronous FIFO using a reference-queue scoreboard and TLM mailboxes.
  – Verified full/empty conditions, read/write sequencing, and data integrity; performing RTL-to-GDSII implementation of the asynchronous FIFO.

Technical Skills

- **HDL and Digital Design**: Verilog, SystemVerilog, RTL Design, FSMs, Sequential Logic, Pipelining
- **Verification**: Self-checking Testbenches, Scoreboards, Constrained-Random Testing, TLM Mailboxes, Waveform Debugging.
- **Programming**: C, Python, MATLAB
- **Tools and Platforms**: Linux, Vivado, Git/GitHub, COMSOL Multiphysics

Positions of Responsibility

- **Founder Community Lead**, Sparkk Club, IIIT Ranchi *Sep. 2023 - May. 2026*
- **Member**, Department Undergraduate Committee, ECE, IIIT Ranchi *Jul. 2024 - Sep. 2025*

Achievements

- **GATE 2026 and 2025**, Qualified; AIR 1109 in Instrumentation Engineering (2026) *Mar. 2026*
- **Academic Excellence**, Ranked 2nd in B.Tech. standings; achieved 10/10 SGPA in Semesters IV–VIII *Feb. 2025*
- **National Space Day 2024**, First Prize for a smartphone-controlled rover with a pan-tilt camera *Aug. 2024*
- **Smart India Hackathon 2023**, 2nd Runner Up, at College stage. *Sep. 2023*